PSet 2: Health Insurance

The scenario

You have applied for a data analyst position at a state agency that advises the legislature on health insurance coverage in the U.S. As a first-round screen, the agency sends every applicant the same dataset and these instructions:

> *Explore the data and choose up to 10 graphs that summarize it. The graphs may not include subtitles, captions, labels, or notes. At the screening, you may bring a printout of your graphs, no other materials. You will be given 30 minutes to answer written questions about the data.*

The agency's analysts are regularly asked questions with no warning, so the screen tests whether you know how to create effective graphs and use them to interpret the data. Expect questions about what your graphs show, why the data look the way they do, and if you have any concerns about the accuracy of the data.

The data

Export data from the U.S. Census by following these steps:

Go to https://data.census.gov/app/mdat, choose "ACS 1-Year Estimates Public Use Microdata Sample" and most recent vintage (should be the defaults) and click Next.

Check the boxes for the following variables `AGEP`, `TEN`, `MAR`, and `HICOV`.

Before downloading the table, we'll need to convert `AGEP` to a categorical variable: click on the cart, choose `AGEP`, click "Create Custom Group", "Auto Group", then set start to 15, end to 99 and groups of 10 and then "Save". Uncheck "Not Elsewhere Grouped" (we won't use the data for ages below 15) and then click "View Table".

Now move all of the variables except `AGEP` into "Rows" to create a long data table. It should look like `Census_data_request.pdf` (Note the options to choose on the left: Count, PUMS person weight and 2024).

Finally, click Download/Share, choose CSV (default) and Export Table Data.

Note: There are some small updates since the last pset in the following sections so please read carefully.

What to do

1. Explore the data thoroughly using techniques from class that are appropriate for this data. Look for patterns as well as unusual features.
2. Choose up to 10 graphs that together give the best summary of the dataset, and which you believe will best enable you to answer potential questions from "the agency."
3. By **Tuesday, November 10, 11:59 pm**, submit a `.qmd` file and rendered `.pdf` file to GradeScope (this assignment).
4. Bring a printout of your graphs to class on **Wednesday, November 11**. During the final 30 minutes of class you will be given questions to answer in writing.

Important: The versions in 3. and 4. should be identical, EXCEPT you should show the code for the version you submit here and hide it in the printout.

To do so, change

```
execute:
  echo: true
```

to `echo: false` in the YAML and render again before you print.

Rules for the graphs

For the purposes of this assignment, the data should speak for themselves. That means:

- Titles may be included to avoid confusion about what's shown in the graph, but should be minimal.
- No subtitles, captions, annotations, or text labels on the plots.
- Axis labels should be minimal. They may be recoded to be more human readable but not contain detailed analyses.
- No added theme elements other than changing the basic theme. For example you can add `+ theme_minimal(12)`.
- Number your figures by adding the figure number as a caption: `+ labs(caption = "Figure 1")`. You may use the figure numbers to refer to specific graphs in your answers to the in-class piece.

Grading

Your grade will be based both on what you turn in to Canvas and your answers to the in-class piece. The assessment will be based on the following components:

Graphs

- Are your graph forms good choices for your data?
- Have you made good choices in parameters?
- In general, have you followed best practices learned in class?

Code

We will not be super picky about your coding style but we do expect you to follow minimal standards:

- Is your workflow reproducible? This is a big topic but for our purposes, it is reproducible if you can answer yes to the following two questions:
  - Would my code run with an updated dataset (say 2025 instead of 2023) (assuming the column names stay the same)?
  - Would my code run on someone else's computer? (No references to your hard drive, etc.)
- Is your code up to date? (No deprecated functions)
- Is your code simple and elegant, within reason?
- Is the code formatted such that it's easy to read and is wrapped so that it is not cut off?

In-class

We will be assessing the following:

- Do you have a good understanding of main trends in the data?
- Do you have a good understanding of oddities, things you don't understand?
- Are you able to recognize questions that cannot be answered by the data and say so?

Tips

- Have a basic understanding of how the data were collected, by whom, and for what purpose.
- Know what each column measures or represents.
- Experiment with your graphs both to understand more about the data and to improve the visualizations. The most value-added comes from this iterative, creative process.
- Color is probably more helpful this time than for PSet 1. I believe there are PAW printers that can do color. Other options are to use pattern instead of color (check out the `ggpattern` extension package.) Or you can manually add color using pens, highlighters, etc. as some of you did last time. I don't think that's as insane/inefficient as it sounds – by looking carefully at the graphs you might pick up some trends or discover things you don't understand.